

SECONDARY FINDINGS 3.3					
Phenotype	Gene	Inheritance	Variants to Report	NOTE	SF 2.0
Genes related to cancer phenotypes					
Familial adenomatous polyposis	<i>APC</i>	AD	All P and LP		Child/adult
Familial medullary thyroid cancer/multiple endocrine neoplasia 2	<i>RET</i>	AD	All P and LP		Child/adult
Hereditary breast and/or ovarian cancer	<i>BRCA1</i>	AD	All P and LP		Adult
	<i>BRCA2</i>				Adult
	<i>PALB2</i>				
Hereditary paraganglioma-pheochromocytoma syndrome	<i>SDHD</i>	AD	All P and LP		Child/adult
	<i>SDHAF2</i>				Child/adult
	<i>SDHC</i>				Child/adult
	<i>SDHB</i>				Child/adult
	<i>MAX</i>				
	<i>TMEM127</i>				
Juvenile polyposis syndrome	<i>BMPR1A</i>	AD	All P and LP		Child/adult
Juvenile polyposis syndrome/hereditary hemorrhagic telangiectasia syndrome	<i>SMAD4</i>	AD	All P and LP		Child/adult
Li-Fraumeni syndrome	<i>TP53</i>	AD	All P and LP		Child/adult
Lynch syndrome (hereditary nonpolyposis colorectal cancer)	<i>MLH1</i>	AD	All P and LP		Adult
	<i>MSH2</i>				Adult
	<i>MSH6</i>				Adult
	<i>PMS2</i>				Adult
Multiple endocrine neoplasia type 1	<i>MEN1</i>	AD	All P and LP		Child/adult

<i>MUTYH</i> -associated polyposis	<i>MUTYH</i>	AR	P and LP (2 variants)		Adult
NF2-related schwannomatosis	<i>NF2</i>	AD	All P and LP		Child/adult
Peutz-Jeghers syndrome	<i>STK11</i>	AD	All P and LP		Child/adult
<i>PTEN</i> hamartoma tumor syndrome	<i>PTEN</i>	AD	All P and LP		Child/adult
Retinoblastoma	<i>RB1</i>	AD	All P and LP		Child
Tuberous sclerosis complex	<i>TSC1</i>	AD	All P and LP		Child
	<i>TSC2</i>				Child
von Hippel-Lindau syndrome	<i>VHL</i>	AD	All P and LP		Child/adult
<i>WT1</i> -related Wilms tumor	<i>WT1</i>	AD	All P and LP		Child
Genes related to cardiovascular phenotypes					
Aortopathies	<i>FBN1</i>	AD	All P and LP		Child/adult
	<i>TGFBR1</i>				Child/adult
	<i>TGFBR2</i>				Child/adult
	<i>SMAD3</i>				Child/adult
	<i>ACTA2</i>				Child/adult
	<i>MYH11</i>				Child/adult
Arrhythmogenic right ventricular cardiomyopathy (a subcategory of arrhythmogenic cardiomyopathy)	<i>PKP2</i>	AD	All P and LP		Child/adult
	<i>DSP</i>			also associated with DCM	Child/adult
	<i>DSC2</i>				Child/adult
	<i>TMEM43</i>				Child/adult
	<i>DSG2</i>				Child/adult
Catecholaminergic polymorphic ventricular tachycardia	<i>RYR2</i>	AD	All P and LP		
	<i>CALM1</i>				
	<i>CALM2</i>				
	<i>CALM3</i>				
	<i>CASQ2</i>	AR			

	<i>TRDN</i>	AR	P and LP (2 variants)		
Catecholaminergic polymorphic ventricular tachycardia & Hypertrophic Cardiomyopathy (HCM)	<i>MYH7</i>	AD	All P and LP		Child/adult
	<i>MYBPC3</i>				Child/adult
	<i>TNNI3</i>				Child/adult
	<i>TPM1</i>				Child/adult
	<i>MYL3</i>				Child/adult
	<i>ACTC1</i>				Child/adult
	<i>PRKAG2</i>				Child/adult
	<i>MYL2</i>				Child/adult
Dilated cardiomyopathy (DCM)	<i>TNNT2</i>	AD	All P and LP		
	<i>LMNA</i>		P and LP	To report: any case level phenotype evidence of association with cardiac disease (eg, DCM, arrhythmogenic right ventricular cardiomyopathy, arrhythmogenic cardiomyopathy, and/or arrhythmia), novel pLOF variants that reach LP without case observations, these variants should be reported given the general association of pLOF LMNA variants with cardiac disease and the evidence summary should include mention of the spectrum of phenotypes that may be observed with LMNA pLOF variation. Do not report: previously reported P/LP missense variants never associated with cardiac disease	
	<i>FLNC</i>		All P and LP	also associated with HCM	

	<i>TTN</i>		P and LP	only frameshift and nonsense variants, and variants known to impact the splicing of TTN exons with high PSI	
	<i>BAG3</i>		All P and LP		
	<i>DES</i>				
	<i>RBM20</i>				
	<i>TNNC1</i>				
	<i>MYH7</i>				
	<i>SCN5A</i>				
	<i>PLN</i>				Also associated with hypertrophic cardiomyopathy (HCM), arrhythmogenic right ventricular cardiomyopathy (ARVC).
Ehlers-Danlos syndrome, vascular type	<i>COL3A1</i>	AD	All P and LP		Child/adult
Familial hypercholesterolemia	<i>LDLR</i>	SD	All P and LP		Child/adult
	<i>APOB</i>	AD			Child/adult
	<i>PCSK9</i>	AD			Child/adult
LQTS types 1 and 2	<i>KCNQ1</i>	AD	All P and LP		Child/adult
	<i>KCNH2</i>				Child/adult
LQTS3; Brugada syndrome	<i>SCN5A</i>	AD	All P and LP		Child/adult
LQTS types 14-16	<i>CALM1</i>	AD	All P and LP		
	<i>CALM2</i>	AD			
	<i>CALM3</i>	AD			
Genes related to inborn errors of metabolism phenotypes					
Biotinidase deficiency	<i>BTD</i>	AR	P and LP (2 variants)		

Fabry disease	<i>GLA</i>	XL	All hemi, het, homozygous P and LP	also applies to the cardiovascular category.	Child/adult
Ornithine transcarbamylase deficiency	<i>OTC</i>	XL	All hemi, het, homozygous P and LP		Newborn (male), child (female)
Pompe disease	<i>GAA</i>	AR	P and LP (2 variants)		
X-linked adrenoleukodystrophy	<i>ABCD1</i>	XL	All hemi, homozygous or 2 het. P and/or LP	A single heterozygous P/LP variant should not be reported.	
Cerebrotendinous xanthomatosis	<i>CYP27A1</i>	AR	P and LP (2 variants)		
Genes related to miscellaneous phenotypes					
Hereditary hemochromatosis	<i>HFE</i>	AR	<i>HFE</i> p.C282Y homozygotes only	NM_000410.3	
Hereditary hemorrhagic telangiectasia	<i>ACVRL1</i>	AD	All P and LP		
	<i>ENG</i>				
Malignant hyperthermia	<i>RYR1</i>	AD	All P and LP	only variant associated with malignant hyperthermia, no neuromuscular phenotype	Child/adult
	<i>CACNA1S</i>				Child/adult
Maturity-onset of diabetes of the young	<i>HNF1A</i>	AD	All P and LP		
<i>RPE65-related retinopathy</i>	<i>RPE65</i>	AR	P and LP (2 variants)		
Wilson disease	<i>ATP7B</i>	AR	P and LP (2 variants)		Child
Hereditary TTR amyloidosis	<i>TTR</i>	AD	All P and LP		